

Supporting Information for:
**Ultrafast Solution-Phase Photochemistry Revealed by Multi-Octave
Continuum Probe**

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A. Details on soliton generation using Ti:Sapphire and Yb:KGW amplified laser systems

1. General details and design principles

To generate a soliton within a practical fiber length ($<3\text{--}4\text{ m}$), it is necessary to either generate a short pulse ($\sim 10\text{ fs}$) or use a small core radius ($< 200\text{ }\mu\text{m}$), see Equation S.1.¹ The latter approach is not recommended since the $1/e$ power loss is proportional to a^3 , thus preventing the use of smaller radii (see Equation S.2). Since soliton formation is highly sensitive to the pulse intensity and duration of the pulse, it is convenient to separate the broadening stage to exert fine control over the pulse that is launched into the soliton generation stage. Current commercial amplifier laser systems (either Ti:Sapphire or Yb:KGW) typically generate pulse widths in the order of $35\text{--}300\text{ fs}$ thus, to facilitate the generation of the soliton we use two stages: 1) compression to $\sim 10\text{ fs}$, and 2) soliton formation. The diagrams of our setups are shown in Figure S.1. and S.2. We used a 1 mJ portion of the 800 nm fundamental of a 1 kHz repetition rate Ti:Sapphire regenerative amplifier (Coherent Legend Elite HE-USP: $\sim 2.7\text{ mJ}$, 29 nm FWHM) with a pulse width of $51 \pm 2\text{ fs}$ FWHM. For the soliton generation in an Yb-based laser system (Light Conversion Pharos) we used the full 1 mJ output centered at 1030 nm at 1 kHz repetition rate and with a pulse width of 184 fs FWHM.²

2. Details on the Ti:Sapphire soliton-probed transient absorption optical setup

The broadening/compressor and soliton stages are shown in Figure S.1. For the Ti:Sapphire system, the compressor stage consists of a fused silica capillary with a length of 150 cm and a hollow core with an inner diameter of $450\text{ }\mu\text{m}$. The fiber is encapsulated in a stainless-steel case that protects and maintains a pressure of 0.2 bar absolute of Argon gas through the fiber. The height of the input beam is brought down by a 90° periscope. The periscope also serves to rotate the polarization from parallel to perpendicular relative to the plane of the table (i.e., rotation from p -

polarized to *s*-polarized with respect to the plane of incidence on the thin film polarizer in the attenuation line, this plane of incidence is parallel to the plane of the table). The input beam can be attenuated using an 800 nm $\lambda/2$ waveplate and thin film polarizer (extinction ratio of 40:1); this allows the reflected part (*s*-polarized) instead of the transmitted part (*p*-polarized), to continue into the focusing line.

The input beam for the compressor stage must be focused such that the $1/e^2$ diameter of the beam waist is 64% of the core diameter to ensure maximum coupling to the EH₁₁ fundamental mode of the fiber.^{3,4} The 800 nm laser beam is focused to a 288 μm $1/e^2$ waist using a 2 m focal length curved mirror. The beam gradually focuses into the fiber entrance as it travels through a 70 cm arm with a 1 mm thick fused silica window. This prevents any nonlinear effects or damage on the window. Argon gas is used as the broadening medium because of its relatively low cost and high nonlinear refractive index. However, it must be used at much low pressures (<1 bar) as to avoid multiphoton ionization, a problem that worsens with the polarizability, *e.g.* as the atomic number increases.^{3,4} These properties allow for sufficient spectral broadening without loss of energy to ionization or the risk associated with using the higher pressures necessary with Helium, or the significant cost of gases such as Neon.

The beam exits through a similar arm, which has a 1 mm thick fused silica output window with a nano-structure treatment to reduce back-reflection over the large spectrally broadened bandwidth (TelAztec FQ25x1-E- RAR-M). The compressor stage has a consistent transmission efficiency of 75%, with the losses attributed to a back-reflection from the front window and refraction losses as the beam reflects at grazing angles through the fiber. Before collimation, the pulse is compressed using a pair of 800 nm chirped mirrors (FewCycle). The chirped mirrors are designed for wavelengths at 500-1000 nm and impart a GVD of $-40 \text{ fs}^2/\text{per pair of reflections}$. In

our configuration, the pulse has a total of 10 pairs of reflections, at an angle of incidence (AOI) of $\sim 8^\circ$, imparting a total GVD = -400 fs^2 . Given the discrete GVD compensation of the chirped mirrors, a pair of fused silica wedges are placed downstream for fine addition of positive GVD arrive at the desired overall compensation. The beam is collimated using a 2.5 m focal length curved mirror to match the appropriate size for proper focusing into the soliton stage.^{3,4}

The beam is attenuated using an achromatic $\lambda/2$ waveplate and a wire grid polarizer. The beam is launched into the soliton stage using a 50 cm focal length curved mirror to target a $96 \text{ }\mu\text{m}$ $1/e^2$ beam waist at the fiber entrance (64% of $150 \text{ }\mu\text{m}$ for optimum coupling to EH_{11} mode). The beam waist measured using the razor blade method is $99 \pm 5 \text{ }\mu\text{m}$, which is at the required size within the uncertainty of the measurement. The beam gradually focuses through a 35 cm arm with a window made of 1 mm thick CaF_2 chosen because of its lower dispersion. The fiber inside the soliton stage has an inner diameter of $150 \text{ }\mu\text{m}$ and a length of 150 cm. The beam exits through another CaF_2 window at the end of a 50 cm arm. The exit arm is longer to ensure the ultrashort, high intensity, and low divergence UV soliton have a larger beam size on the exit window. This is to avoid damage to the exit window, which has been reported in prior studies using shorter exit arms.¹

More details regarding the transient absorption setup are provided in section B.

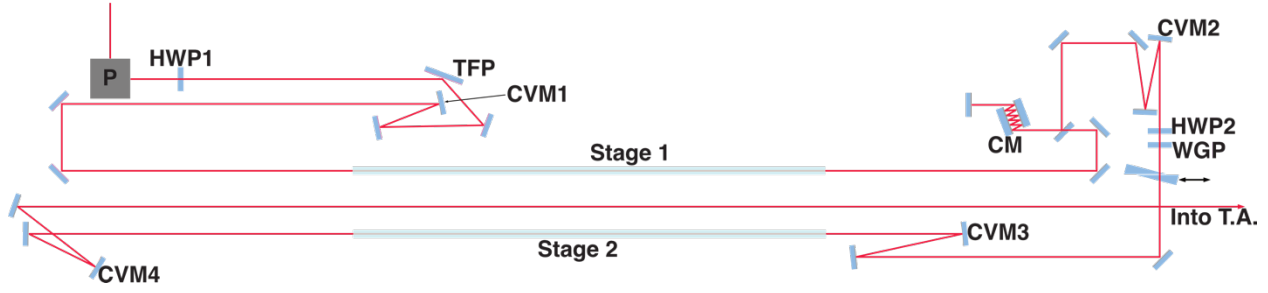


Figure S.1 Setup for soliton generation in a Ti:Sapphire system. Not to scale. Compressor stage (1) and soliton stage (2). Optical components for compressor stage: (P) Periscope, (HWP1) 1/2 waveplate, (TFP) thin film polarizer, (CVM1) 2.0 m focal length concave curved mirror, (CM) 800 nm, (CVM2) 2.5 m focal length concave curved mirror, (HWP2) achromatic 1/2 waveplate, (WGP) wire grid polarizer. A pair of wedges is used for GDD compensation. The first wedge is bolted to the table at a position to minimize the amount of glass transmitted through and the second wedge is on a micrometer for fine tuning. Optical components for soliton stage: (CVM3) 50 cm focal length concave curved mirror, (CVM4) 75 cm focal length concave curved mirror.

3. Details on the Yb:KGW soliton-probed transient absorption optical setup

The design for soliton supercontinuum generation using a Yb:KGW laser amplifier is similar to the system based around a Ti:Sapphire amplifier. Since the fundamental pulse duration is longer, the compression from 200 fs to 10 fs was accomplished using two sequential stretched hollow-core fiber stages, each accompanied by a chirped mirror compressor. The first compressor stage consists of a hollow-core fiber with an inner diameter of 530 μm , a length of 90 cm, and filled with Argon gas at 1.8 bar absolute pressure. The output is compressed using a chirped mirror compressor down to ~ 30 fs (Ultrafast Innovations, HD120, 5° AOI, -200 fs^2). The second compressor stage takes a 90% portion of the 30-fs pulse (10% used for UV generation using fourth-harmonic generation). This second fiber has the same inner diameter but is filled with 0.4 bar absolute of Argon gas and is 180 cm long. The output from this second compressor stage is compressed with chirped mirrors down to ~ 10 fs (UltraFast Innovations, PC147, 5° AOI, -60 fs^2).

The transmission of the first compressor stage in the Yb system was consistently 84% while the second stage had a transmission of 71%. As just described, the losses are attributed to refraction as the beam bounces inside the capillary. The major difference between the optical setups used in the Ti:Sapphire system and the Yb-based system is the use of lenses for focusing and collimating the beam into the fiber. This choice is based on the much longer starting pulse used in the Yb system, which is not affected significantly by the GVD of the optics. However, working at high pulse energies produced some noticeable self-phase modulation (from transmission through lens material) in the input pulse, but it appears as if this had no great effect in the performance of the system.

The compressed output is then directed through an attenuation line, composed of a $\lambda/2$ achromatic waveplate and a wire grid polarizer, followed by a pair of fused silica wedges for GVD compensation and finally focused into the third fiber using a 50 cm focal length curved mirror. The soliton continuum is generated in a third stage consisting of a 90 cm long fiber with an inner diameter of 200 μm and filled with Neon at 1.4 bar absolute pressure.

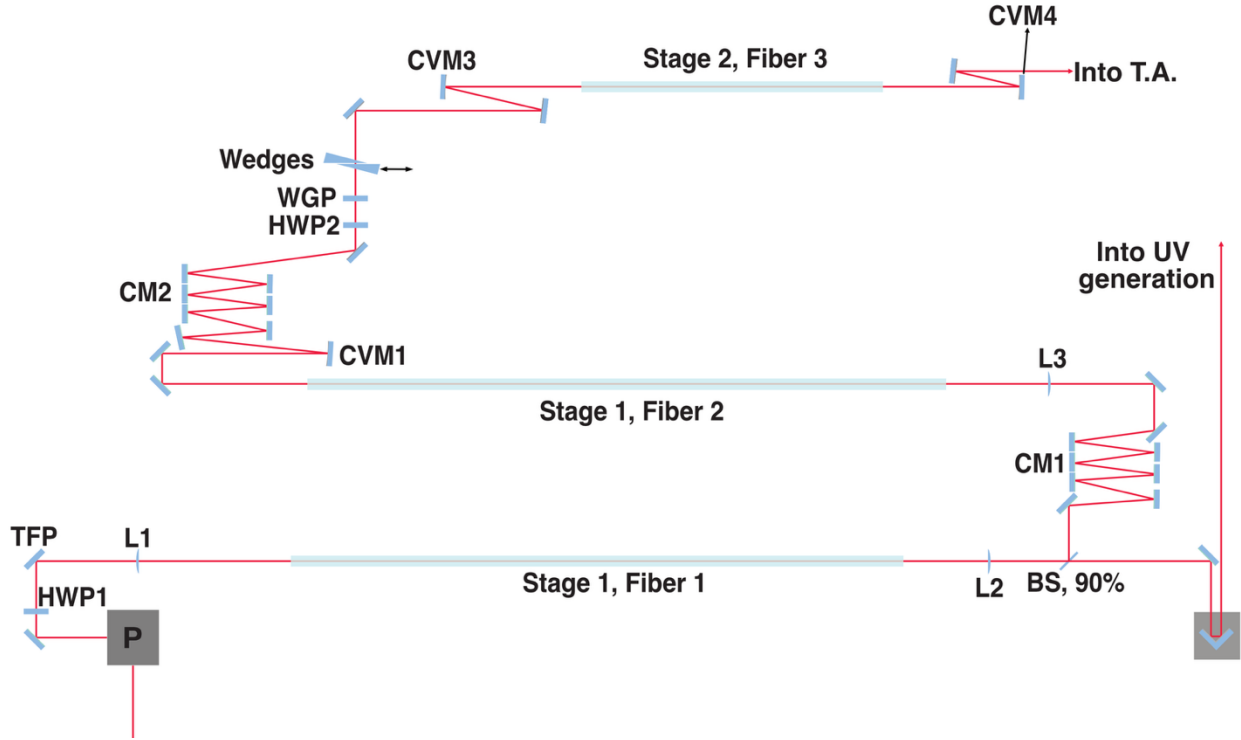


Figure S.2. Setup for soliton generation in an Yb:KGW system. Not to scale. Compressor stage (fiber 1 and 2) and soliton stage (fiber 3). Optical components for compressor stage: (P) Periscope, (HWP1) $\lambda/2$ waveplate, (TFP) thin film polarizer, (L1, L2, and L3) 2 m focal length convex lens, 90% beam splitter (BS), 950-1120 nm chirped mirrors (CM1), concave curved mirror 2 m focal length (CVM1), 650-1350 nm chirped mirrors (CM2), (HWP2) achromatic $\lambda/2$ waveplate, (WGP) wire grid polarizer. A pair of wedges is used for GDD compensation. Optical components for soliton stage: (CVM3 and CVM4) 50 cm focal length concave curved mirrors, off-axis parabolic mirrors for focusing into the sample (101.6 mm effective focal length), collimating after the sample (50.8 mm effective focal length), and coupling into the spectrometer (203.2 mm).

4. Soliton fission length tuning

A soliton spanning from UV to NIR can be achieved by tuning the conditions that favor an intense soliton spectrum. Doing so requires tuning the fission length (L_{Fission}) of the system.

Equation S.1 shows the variables on which the L_{Fission} depends:⁵⁻⁷

$$L_{\text{Fission}} \propto \frac{\tau_{\text{FWHM}} a^2}{\sqrt{I_0}} \quad \text{Equation S. 1}$$

Equation S.1 indicates that to achieve a practical fission length it is desirable to have a short pulse (τ_{FWHM}) and intense pulse (I_0), or small inner core radius (a). Although the square dependence of a suggests that using smaller radii would support a practical fission length, the $1/e$ power loss length (L_{Loss}), that scales proportional to a^3 , significantly reduces the intensity of the pulse as travels through the fiber (Equation S.2).

$$L_{Loss} \propto \frac{a^3}{\lambda^2} \quad \text{Equation S. 2.}$$

where λ is the wavelength. Therefore, the variables that are easier to adjust are the pulse width and the input intensity. To demonstrate the significance of $L_{Fission}$ to the task of soliton tuning, two Luna simulations were performed. The simulation conducted in Luna used an 800 nm, 50 fs, 1 mJ pulse input into a 450 μm core diameter and fiber length of 150 cm filled with 0.2 bar of Ar. The simulated pulse is then compressed by propagation through 10 pairs of PC70 chirped mirrors (UltraFast Innovations) followed by GDD tuning using different amounts of fused silica to lengthen/shorten the pulse. Figure S.3 (a, b) shows a simulation of the spectral (left, horizontal axis) and temporal (right, horizontal axis) evolution over the propagation length of the fiber (vertical axis). When the pulse is optimally compressed (propagation through 10 mm of fused silica), the fission length occurs at ~ 75 cm (red, dashed line). At the fission length when the higher order dispersion becomes non-negligible, the superposition of fundamental solitons breaks apart. This is observed in the spectral domain by a large spectral broadening into the deep UV followed by an abrupt spectral narrowing as seen in Figure S.3. (a) around ~ 80 cm. In the time domain, we see in the simulations a shortening of the pulse width and a subsequent splitting of the pulse into multiple solitons (seen as ripples indicated by the orange arrow).⁵⁻⁷ Therefore, it is necessary to tune the system conditions so that the fission length is approximately the same as the length of the

fiber to maintain a significant spectral energy density throughout the spectrum at the moment the pulse exits the fiber.^{1,5} This is exemplified by a second simulation (Figure S.3 (b)) where the pulse is propagated through 17.8 mm of fused silica to obtain a longer pulse width, making the fission length much longer and favoring a soliton with significant spectral intensity in the visible and UV. These new conditions set the fission length farther down the fiber at ~140 cm, see Figure S.3 (b).

Figure S.4 and S.5 show the comparison of the spectral outputs of these simulations and experimental measurements—it is clear when the pulse is longer the spectral energy density in regions between 210-700 nm is higher. A similar experiment was carried out using the soliton output generated in Yb:KGW, where instead of changing the pulse width by adding dispersive material (as in the case of Ti:Sapphire), we changed the input pulse intensity. The results of these experiments, shown in Figure S.6, show a similar trend as in the case of the Ti:Sapphire where the intensity increases in the UV and visible regions as the fission length is increased by changing the input pulse intensity. The region 200-320 nm appears modulated because of non-phase matched third harmonic generation that occurs inside the fiber. This was observed as large amounts of noise in this region in our time-resolved spectra, especially near 300 nm, that was somewhat improved by a higher number of scans.

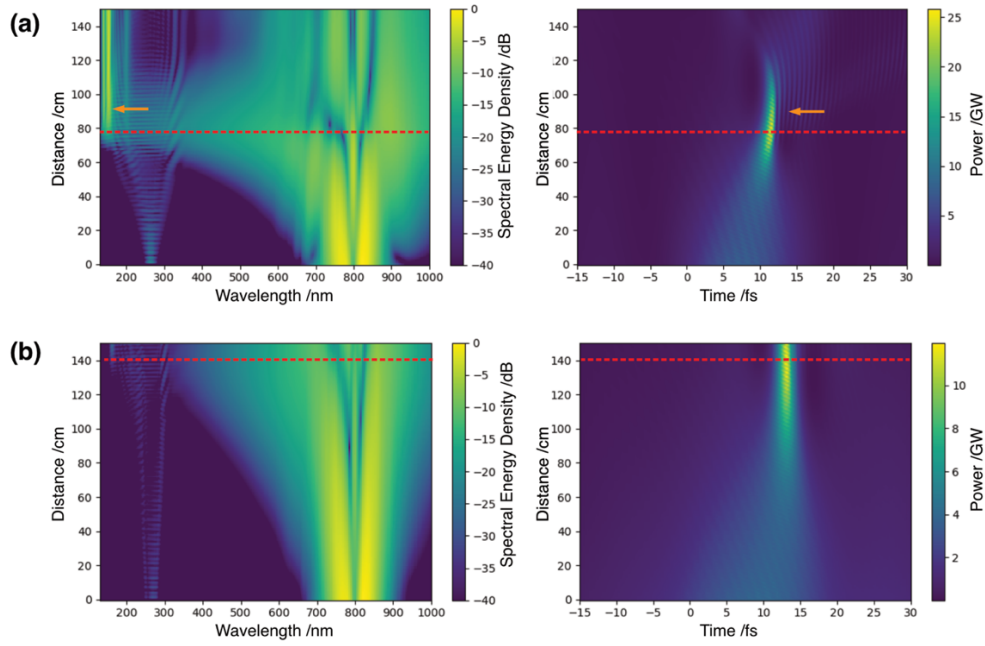


Figure S.3 Simulation of the propagation of a pulse into the soliton stage originating from a Ti:Sapphire system. The energy of the input pulse was 120 μJ and the pressure is 2.0 bar absolute. (a) Propagation of the pulse after addition of 10 mm of fused silica for optimal compression at 12.3 fs. The fission length (red, dashed line) occurs 70 cm away from the fiber exit, thus favoring the formation of a resonant dispersive wave (RDW, orange arrow).¹ (b) Propagation of the pulse after addition of 17.8 mm of fused silica to increase the pulse width to 19.2 fs. This longer pulse width pushes the fission length to the end of the fiber.

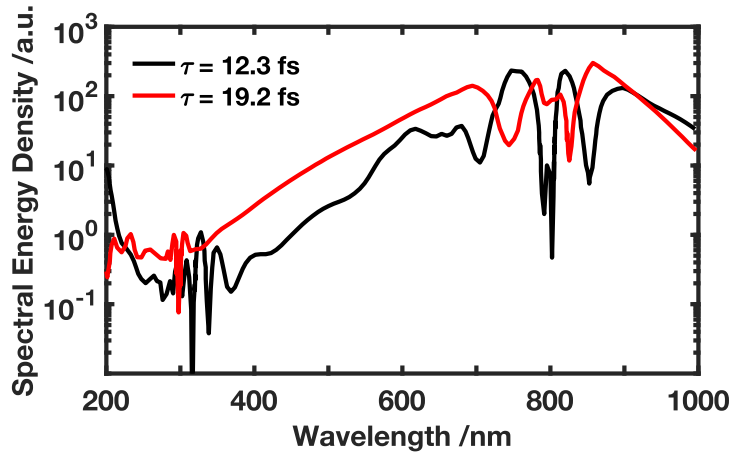


Figure S.4 Simulated spectral output for the simulations described in Figure. S.3. The pulse width of an input pulse was altered by increasing the amount of fused silica the pulse propagates through from 10 mm to 17.8 mm (black and red lines, respectively). These changes solely in pulse width changed the spectral output, increasing the spectral energy density in regions from the visible to the deep UV.

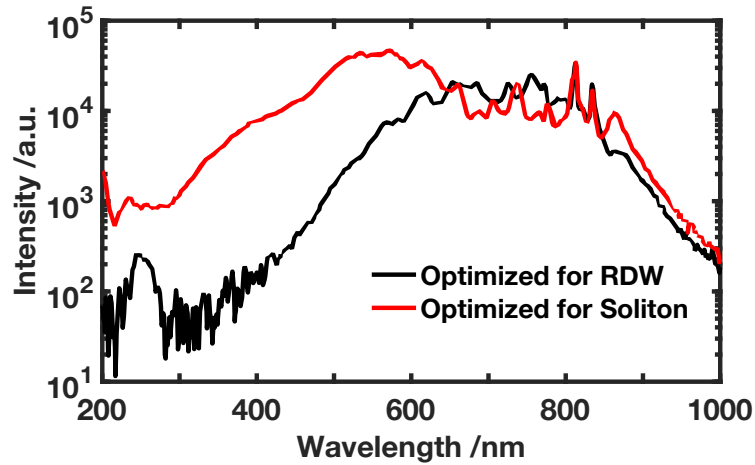


Figure S.5 Experimental spectral outputs of the soliton stage with an input of $\sim 370 \mu\text{J}$ and a pressure of 1.9 bar abs. of He. The spectrum in black was collected when the pulse compression was optimized. Consecutively, fused silica was added using a pair of wedges to optimize the soliton emission. Note: The spectral output from this figure and Figure S.4 are quite different because: 1) the conditions for simulation and experiment were close but not identical and 2) the experimental spectra collected are not corrected to consider the spectrometer's sensitivity at different wavelengths.

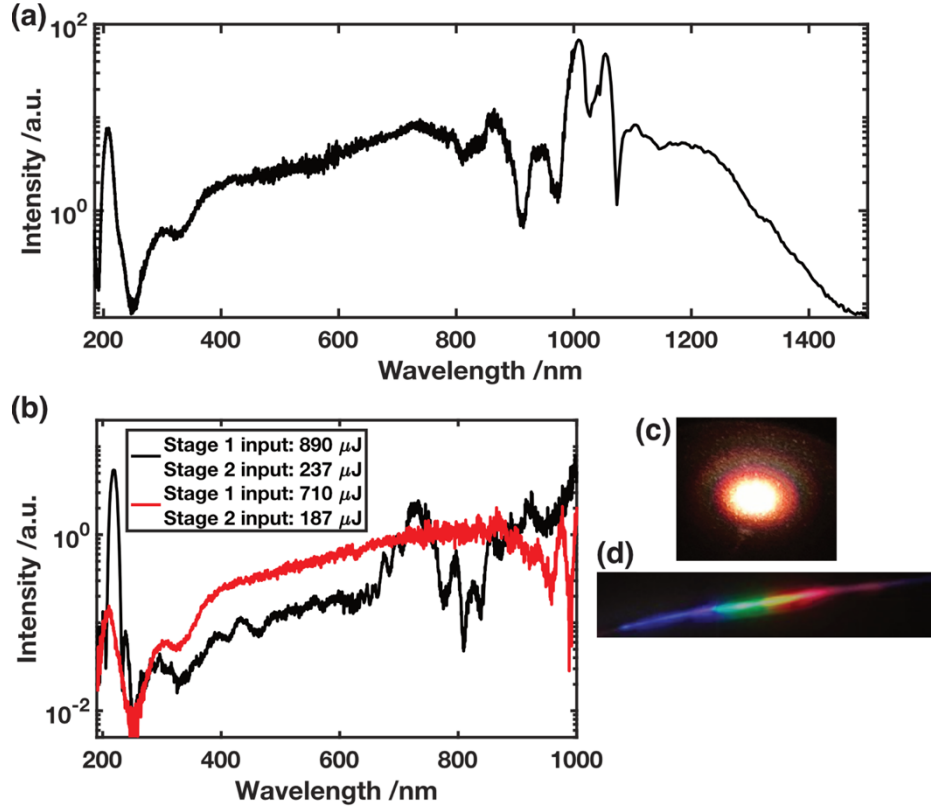


Figure S.6 Spectral output of the soliton generation stage using a Yb:KGW laser system. (a) Full spectral extent of the soliton generated using 189 μ J of the compressed output generated in the previous compressor stages. The gas inside of the fiber is Ne at 1.4 bar abs. This continuum spans from 200-1400 nm and with a total output of 60 μ J more than any bulk material has achieved. (b) The soliton continuum can be tuned to obtain the best conditions where the highest energy density in the continuum (red) is achieved instead of in the RDW (black). This is done by increasing the fission length by reducing the energy input into the first compressor stage. This will effectively change the pulse width and the pulse intensity. (c) The output profile of the soliton generation stage has a nice Gaussian shape that helps tightly focusing the beam into a sample at the interaction region. (d) The spectrum of the soliton generation shown dispersed by a grating. It shows its extent from the UV (left, blue fluorescence from card) to the red (NIR not captured).

B. Design and implementation of a prism spectrometer required for use of a soliton continuum for multi-octave transient absorption

A prism spectrometer has a primary advantage over a grating spectrometer in that it does not suffer the problem of higher order diffraction overlapping with the $n = 1$ order. Instead of diffracting the light (which causes the problem of overlapping diffractions) as a grating would, a prism disperses the transmitted light at different angles as function of the wavelength because of the wavelength variation of the refractive index.

The spectrometer design used for our electronic transient absorption experiments is shown in Figure S.7. and S.8. The light is focused into the spectrometer using a 1" diameter, 15 cm focal length curved mirror (not shown) and the background light gets filtered out with an adjustable slit. The beam is collimated using another 1" diameter, 15 cm focal length curved mirror (CVM B). The light reflected off the collimating mirror arrives at Brewster's angle to a CaF_2 prism with an apex angle of 60° . Brewster's angle was set by sending the outgoing beam for ~ 40 cm and finding the angle of minimum deviation of most of the visible components of the beam. The outgoing beam is directed with a 1" diameter flat mirror onto a 2" diameter flat mirror. The 2" diameter mirrors are used because of the increasingly large size of the beam as the light disperses. After the previously mentioned filter, the light is selectively attenuated with a UV fused silica continuous linear ND filter (increasing from 0.04 to 2.0 OD, Thorlabs NDL-25C-2) where the deep UV wavelengths will undergo the least attenuation and the NIR wavelengths will undergo the most attenuation. The dispersed beam is then focused into the camera using a 2" diameter, 15 cm focal length curved mirror. The camera used is a 1 kHz Entwicklungsbüro EBStresing camera with a Hamamatsu back-thinned 1024-element CCD (S7034-1007S). The CCD has a quartz window (no

coating) with > 90% transmission and a spectral response of > 40% quantum efficiency from 200-950 nm.

Regardless of the advantages of the soliton probe over the spectra generated from other sources in bulk media, there are still important considerations that must be taken to correctly use this light source. The first consideration is the significantly larger pulse energy that is carried within the soliton output. In both systems that have been described here, the pulse energy is far above the typical < 2 μ J used in bulk-media-probed transient absorption. Therefore, to avoid damaging, heating, or other nefarious changes induced by the high intensity of the probe, appropriate attenuation must be done before probing a liquid sample. However, this is easier said than done since the amount of light in the UV is much lower than the visible and NIR; thus, using regular ND filters (neutral density) will not solve the problem completely since they will attenuate every wavelength almost uniformly. Low-pass filters or filters that allow the UV to pass but only attenuate the visible or NIR (*e.g.*, UG filters) are hard to come by and often are limiting in how much bandwidth in the UV and visible they transmit. Fortunately, other researchers^{8,9} in the field have come up with methods that effectively allow for selective reflectivity between 200-400 nm while greatly reducing the reflectivity of wavelengths between 400 nm into the NIR. This is done with a system of Si plates (Crystran) set to reflect near or at Brewster's angle. In our setup shown in Figure S.8 the first two Si plates are oriented with their surface at $\sim 50^\circ$ with respect to the plane of the table and with the incident beam's polarization perpendicular to the plane of the table (*i.e.*, *p* polarized to the plane of reflection of the two Si plates). Setting the plates like this allows for the wavelengths in the UV and blue to have higher reflectivity while wavelengths above 400 nm and towards the NIR to be attenuated. The last Si plate's surface is oriented perpendicular to the plane of the table so that the polarization of the incident beam is *s*, thus having high reflectivity over all

wavelengths, but low reflectivity of p polarized light. This step helps increase the polarization purity coming out the fibers. This is recommended since it has been observed that the polarization purity decreases after passing through a fiber.⁹ This is attributed to deviations from the cylindrical shape of the inner core of the fiber which induces coupling of a fiber mode's polarization to a mode with orthogonal polarization.^{5,9} The effect is the same whether the discriminating plates or polarization-cleaning plate are set in either order - the order is subject to convenience. The overall effect is a beam with clean polarization and a greatly attenuated visible and NIR region while the UV is better preserved.

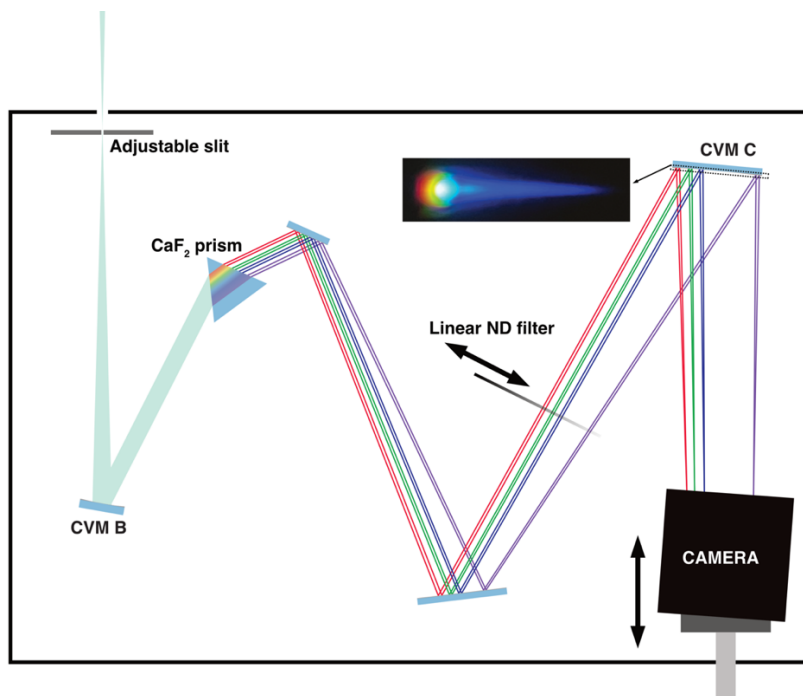


Figure S.7 Diagram of prism spectrometer used for soliton-probed transient absorption. The light is focused into the spectrometer through a slit by a 15 cm focal length curved mirror. The light is collimated by an identical curved mirror (CVM B) and directed into a CaF₂ prism which disperses the light. The dispersed light is further selectively attenuated by a linear ND filter placed on a slider and oriented as to attenuate the longer wavelengths more than the UV. The dispersed and attenuated spectrum (see the inset image) is focused with a 15 cm focal length curved mirror into the camera placed on a micrometer. The micrometer allows the camera to be moved into the focus of the beam. The camera is tilted to compensate for the slightly different distances travelled by different colors of the beam, thus allowing the different colors to focus on the same plane. This could ideally be fixed with a different geometry, but the space for the spectrometer was limited.

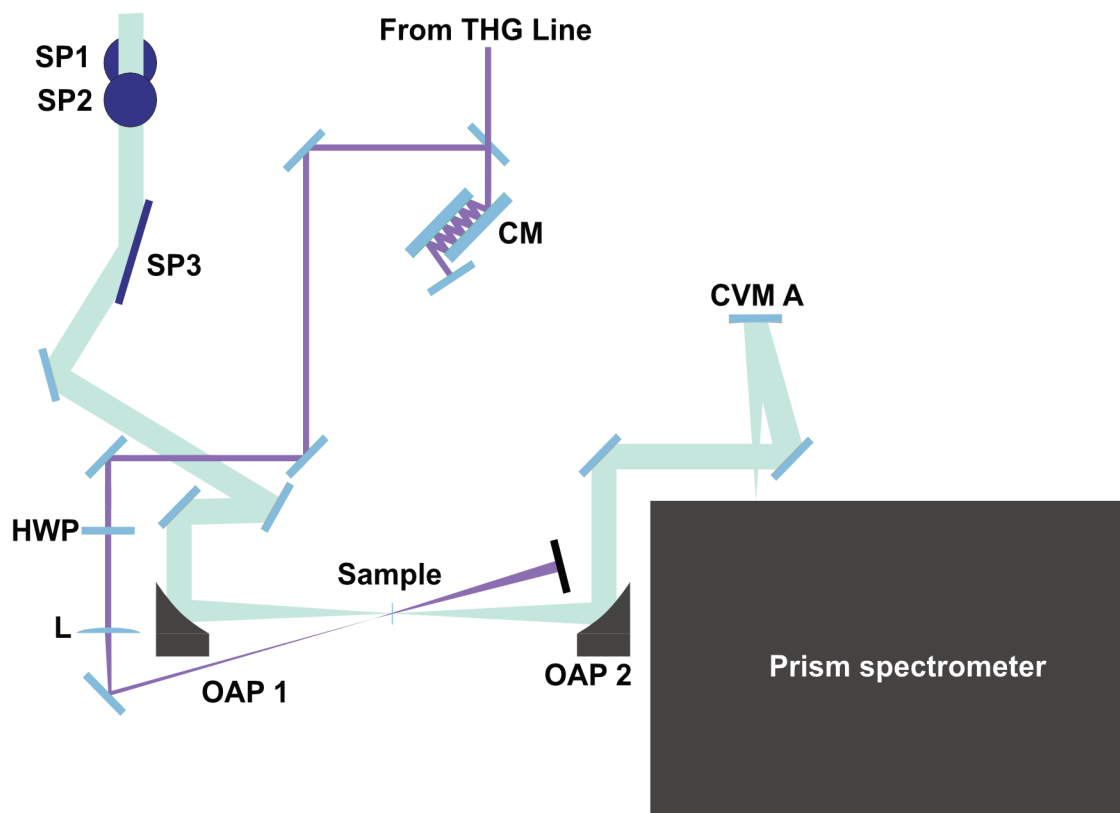


Figure S.8 Diagram for the UV-pumped, soliton-probed transient absorption experiment. The soliton probe is attenuated and its polarization cleaned using a set of three Si (2" diameter) plates set at $\sim 50^\circ$ (SP1,2) and Brewster's angle $\sim 74^\circ$ (SP3), respectively. The probe is focused into the sample and collimated after the sample using a pair of identical off-axis parabolic mirrors (OAP1, 2). The light is then directed and focused into the prism spectrometer using a 15 cm focal length curved mirror (CVM A). The pump is generated through an existing third harmonic line after delaying the fundamental with respect to the probe. The UV pump is compressed using a pair of Gires-Tournois mirrors (CM).¹⁰ The polarization of the UV pump is set with a 267 nm $\lambda/2$ waveplate (HWP) and then focused into the sample with a 25 cm focal length CaF_2 lens (L).

Calibration of the prism spectrometer is challenging due to the large laser bandwidth spanning from the NIR to the deep-UV. The spectrometer calibration is carried out using a Hg (Ne) lamp (Pen ray from UVP, P/N 90-0016-01), a set of visible and NIR filters, and a solution of 6.25 mM benzene in cyclohexane, see spectra on Figure S.9 (a). These three methods provide two

sources of energy resolved, well characterized reference points which allow for calibration down to 5.6 eV.

Once the energy points are collected at a given pixel, we use a simplified (Equation S.4) form of the McKinney-Friedel equation (Equation S.3) for prism spectrometers (below are shown the version originally modified for IR-UV and our simplified form for visible-UV regions that does not include the contribution of the IR region) to create a calibration from pixel to energy:¹¹

$$P(E) = \frac{A}{E_{UV}^2 - E^2} + \frac{B}{E_{IR}^2 - E^2} - P_0 \quad \text{Equation S. 3}$$

$$P_{\text{simp}}(E) \cong \frac{A}{E_{UV}^2 - E^2} - P_0 \quad \text{Equation S. 4}$$

$P(E)$ is the pixel position as a function of energy. The A and B parameters determine the magnitude of the correction in the UV and IR region, respectively. The parameters E_{UV} and E_{IR} represent characteristic absorption energies of the prism material. The parameter P_0 is an offset based on the location where the light is on the camera pixel array. Our simplified form removes the parameters B and E_{IR} since the soliton spectrum does not span into the IR and it was observed experimentally to not make a difference. A fit of the equation to all the calibration points is shown in Figure S.9 (b). The fit shows that the selected points span most of the soliton spectrum and fit well to the McKinney-Friedel equation.

The prism spectrometer resolution was determined using the known width of one of the Hg lamp emission lines at 3.06 eV (404.6 nm). The FWHM at this line in our spectrometer is 0.04 ± 0.01 eV (or 4 ± 3 nm). For reference, the width of this line is 0.02 nm when measured in an instrument with a spectral resolution of 0.016 nm (even when not fully resolved).¹² Nevertheless,

the resolution of our spectrometer is sufficient for most electronic transitions characterized by broad features.

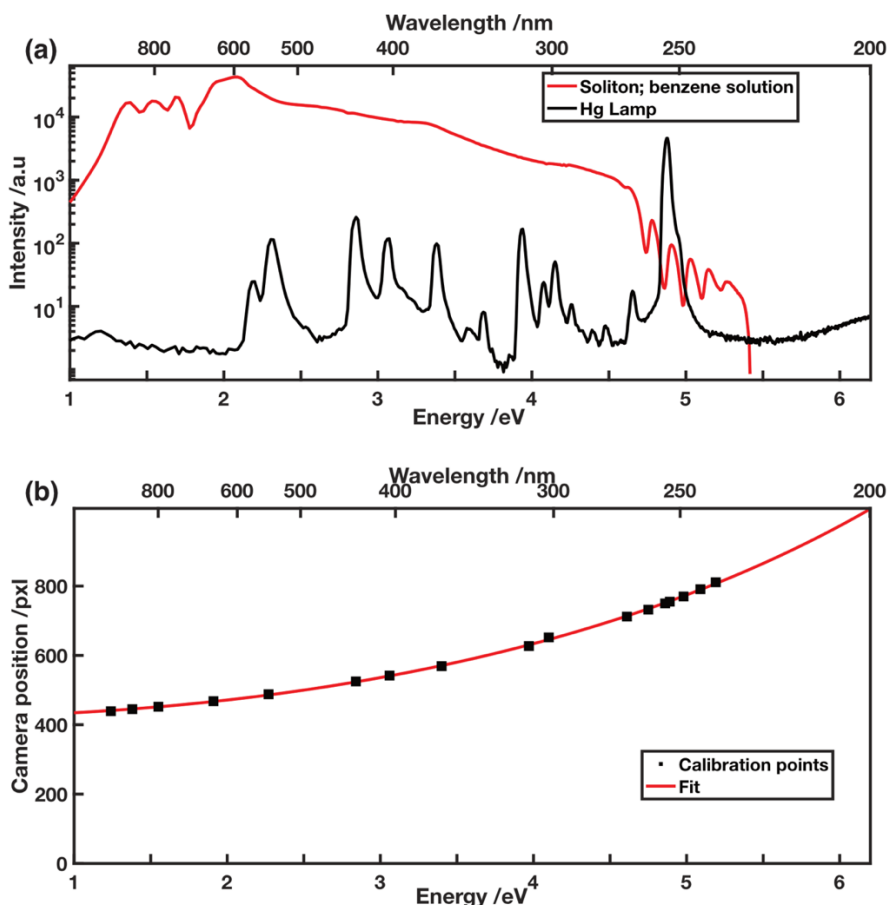


Figure S.9 The energy calibration process for a soliton probe is done using two sets of well-known standards: a Hg lamp and a benzene solution in cyclohexane. The points are then fit to Equation S.4 to create a calibration from camera position to energy in eV. (a) In red is the soliton spectrum collected at the camera after transmission through a 4 mm quartz cuvette with a solution of benzene in cyclohexane. The sharp bands between 4.5 and 5.5 eV are well known absorption bands of benzene - these dips are used as calibration points in the deep UV. In black is a Hg lamp emission spectrum. The most prominent peaks are selected as calibration points for the visible and the UV. (b) The selected points from the Hg lamp or the benzene solution are used to fit the modified form of the McKinney-Friedel equation which translates pixel to energy.

C. Comparison of transient absorption spectra collected using a soliton probe in Ti:Sapphire and Yb:KGW systems.

In the main text, we demonstrated the generation of soliton probes in both Ti:Sapphire and Yb:KGW, but focused on showing the broadband transient absorption implementation using the Ti:Sapphire system. However, transient absorption data for the solvated electron photoproduct generated after UV two-photon ionization of water was collected with both laser systems. Figure S.10 compares the solvated electron transient signature recorded for $t > 3.5$ ps obtained from both systems. For the experiment performed using the Ti:Sapphire amplifier, we used 1 μ J 267 nm with a spot size diameter of 100 μ m FWHM. For the experiment performed using the Yb:KGW system, we used a 0.7 μ J of 257 nm with a spot size of diameter 90 μ m FWHM. The spectra show different spectral regions because the spectrometers and cameras were not set up to capture the same regions. Both spectra show good agreement with the absorption spectrum of the solvated electron.^{13,14} In the case of the Ti:Sapphire probed spectrum, an extra photo-induced absorptive feature appears at probe energies > 3.8 eV is attributed to the solvated OH radical.^{13,15,16}

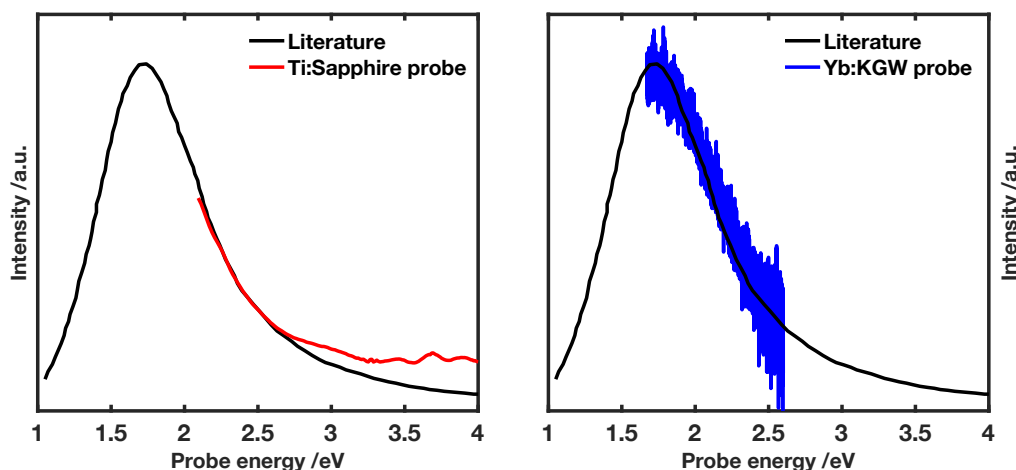


Figure S.10 Transient absorption spectra of the solvated electron in water. The literature spectrum used for the comparison is from Thomsen et al.¹³ (Left) Transient absorption spectra obtained after a two-photon ionization of water with a 267 nm pulse and probed with a soliton continuum

generated from a Ti:Sapphire amplifier source. (Right) Transient absorption spectra obtained after a two-photon ionization of water with a 257 nm pulse and probed with a soliton continuum generated from a Yb:KGW amplifier source. The data collected with the Yb:KGW system shows considerably higher noise due to incomplete optimization of the supercontinuum stability.

D. Sample details

HPLC grade water was used for water 2PA measurements, and the sample was delivered in a 100 μm thick wire-guided gravity jet. We used 15 nJ of 267 nm focused to 100 μm (FWHM) spot size to pump the sample, ensuring no solvated electrons were generated and minimizing cross-phase modulation. For the dimethyluracil photochemistry experiment, DMU was purchased from TCI (98% purity) and used without purification and dissolved in HPLC grade water to make ~ 10 mM solutions. TA data were acquired in a 120 μm liquid jet, which at the 267 nm pump excitation wavelength corresponded to an OD ~ 1.1 . 550 nJ of 267 nm were used to excite the sample, and the pump spot size at the focus was 170 μm (FWHM).

E. Modelling the spectral contribution of hot ground state and ground state bleach in the photophysics of DMU

The overlapping nature of the photohydrate, ground state bleach, and hot ground state spectra make it difficult to deconvolve the contributions of each in the time resolved spectra. For this reason, we provide a very simple simulation that shows the spectral evolution of the hot ground state/ground state bleach and the emergence of the photoproduct. This analysis is essential to calculate the quantum yield of photohydrate formation (see the next section).

Our simulation makes use of the temperature-dependent absorption spectra of DMU in water recorded on a Cary 60 UV/vis spectrometer with a temperature jacketed cuvette compartment; these are shown in Figure S.11. The normalized spectra clearly show that with increasing temperature, there is a significant broadening on the red side of the bands centered at 4.67 eV and 6.08 eV; on the other hand, the blue side of the band centered at 4.67 eV shows negligible broadening. Using these spectra, we have empirically parameterized the asymmetrically broadened spectrum with a temperature dependent peak half width. In essence, the spectrum is divided into three sections: 1) band centered at 6.08 eV with spectral extent of $E < 6.08$ eV; 2) blue side of band centered at 4.67 eV with spectral extent of $E > 4.67$ eV; 3) red side of band centered at 4.67 eV with spectral extent of $E < 4.67$ eV. The bands are represented simply by a sum of Gaussian functions.

We then treat the cooling simply in terms of a time dependent characteristic temperature. The cooling is considered as sequentially redistribution of energy within the DMU molecule and the immediate solvent sphere and then the dissipation of the initially localized heat into the larger solvent volume. This sequential cooling is represented with a biexponential function:

$$T(t) = 293 \text{ K} + \Delta T_{\text{internal}} \exp\left(-\frac{t}{\tau_{\text{internal}}}\right) + \Delta T_{\text{external}} \exp\left(-\frac{t}{\tau_{\text{external}}}\right) \quad \text{Equation S. 5}$$

Here, the initial change in temperature of DMU after absorption of a photon and before the heat is transferred to the bath is $\Delta T_{\text{internal}}$; this temperature change cools with a time constant τ_{internal} . When the heat is transferred to the bath the temperature changes experienced by the molecule solvated in this hot bath is given by $\Delta T_{\text{external}}$ and it cools with a time constant τ_{external} . The latter phase is reasonable based on use of UV/vis spectra of elevated temperature solutions, whereas a more complex description of the intramolecular vibrational redistribution stage and how that

affects the absorption spectrum is clearly warranted¹⁷, but beyond the scope of our current work. Importantly, to include the contribution of the photohydrate and the unrecovered bleach due to the reacted molecules, we add an offset of the spectrum formed by the photohydrate and the simulated ground state spectrum of DMU at 293 K.

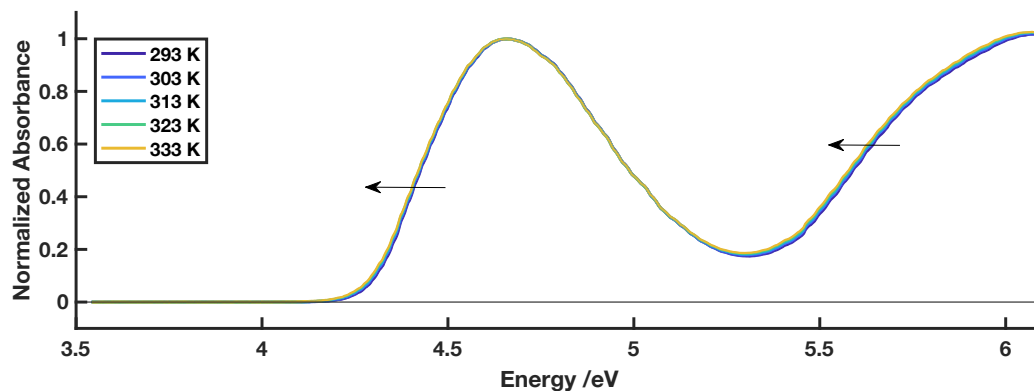


Figure S.11. Spectra of DMU dissolved in water collected at different temperatures. The increase in temperature causes an asymmetric broadening of the two observable bands; arrows indicate direction of broadening as temperature increases.

Based on this simplified description of the overall cooling processes, we present the simulated transient absorption spectrum with (Figure S.12 (a)), and without contributions from the photohydrate (Figure S.12 (c)), and the experimental spectrum (Figure S.12 (b)). The parameters used to reach agreement between the simulation and experimental spectra are as follows: $\Delta T_{\text{internal}}$ is 3000 K, $\Delta T_{\text{external}}$ is 200 K, τ_{internal} is 3 ps, and τ_{external} is 15 ps. The initial selection of the temperature jump in these parameters were based on work in uracil and thymidine, which provide estimates for the internal temperature after excitation at ~ 1200 K and 1800 K, respectively.^{18,19} The external temperature difference was selected to best match the observed behavior. A significant discrepancy between the previously reported temperature changes and

those ultimately selected here may originate in the assumption that broadening is linear with temperature, which is known to not be the case.²⁰

From the comparison of the simulated spectra and the experimental spectra we can see the hot ground state absorption and ground state bleach give a characteristic dispersive-like transient signature at DUV probe energies. This feature evolves spectrally as its amplitude diminishes when hot molecules cool to the ground state, allowing for the emergence of the photohydrate and a matching bleach amplitude that does not recover because of the formation of a new photoproduct. When the photohydrate contribution is artificially removed (Figure S.12 (c)) the cooling (and bleach refilling) is complete, and thus its contribution to the simulated TA spectrum at 520 ps is negligible, indicating that the remaining *experimental* signal in the DUV at 520 ps corresponds solely to the photohydrate and the unrecovered bleach. As we have isolated the origin of the long-time signal, we can calculate the quantum yield for photohydrate formation using our TA data.

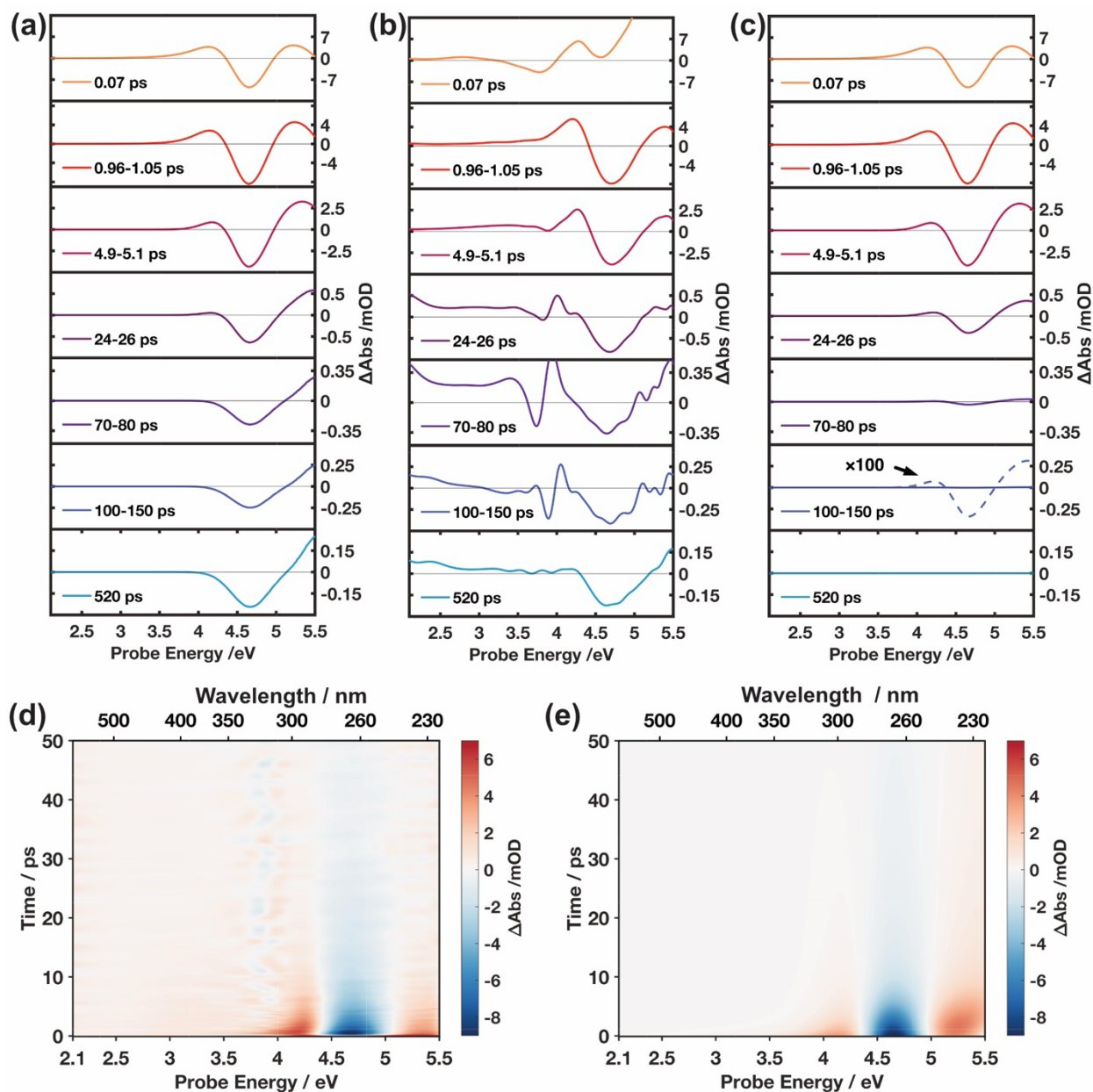


Figure S.12. Simulated spectra of DMU. The simulation does not include the spectral contributions of the band at 2.5 eV attributed to the triplet or the excited state absorption and stimulated emission observed at 0.07 ps. (a) Simulated spectra with the photohydrate contribution included, (b) Tukey-filtered experimental spectra – see below, and (c) simulated spectra without the photohydrate contribution. The dashed line at (c) is the simulated residual signal ($\times 100$) demonstrates the hot ground state absorption and associated ground state bleach are negligible by 100-150 ps. (d, e) The full TA spectrum and the simulated spectrum, respectively, show similar spectral evolution over the first 50 ps.

F. Photohydration quantum yield calculation

Our calculation starts by determining the concentration of photohydrate generated after excitation. The residual signal obtained at 520 ps (solid, violet line) is decomposed into a linear combination of the absorbance spectra of bleached DMU (dashed, black) and photohydrate²¹ (dashed, green); the amplitudes of the basis spectra are then used to reconstruct the TA signal (dotted, red line) as shown in Figure S.13.

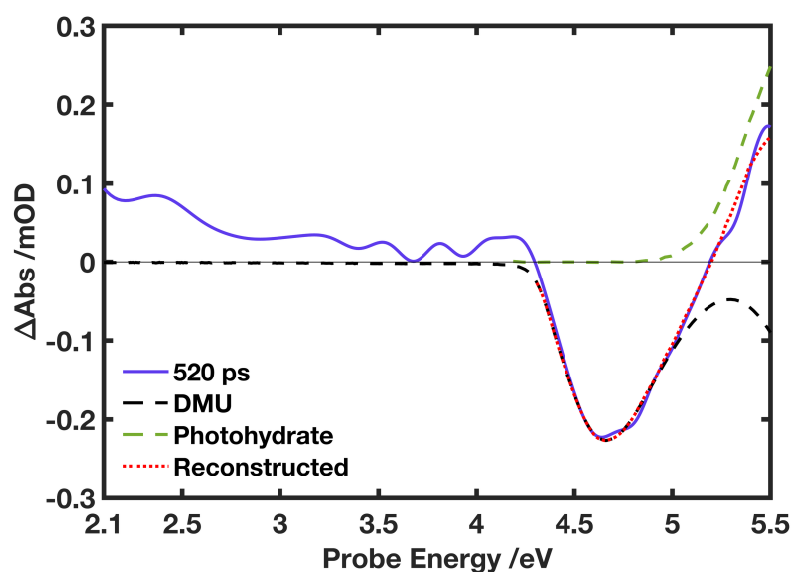


Figure S.13 Tukey-filtered experimental spectra (see below for details) used for the determination of the concentration of DMU photohydrate.

The absorption of the photohydrate was obtained from the amplitudes determined in this reconstruction ($A_{226\text{ nm}} = 2.4 \times 10^{-4}$). In order to obtain a concentration of the photohydrate, we calculated the molar absorption coefficient based on the absorption spectrum of the isolated photohydrate of DMU presented in Moore *et al.*, $\epsilon = 4,975\text{ cm}^{-1}\text{ M}^{-1}$ at 226 nm.²¹ While the authors did not directly report an extinction coefficient, we compared their spectrum of DMU, which has a known molar absorption coefficient ($8,900\text{ cm}^{-1}\text{ M}^{-1}$ at 267 nm), and determined the

path length these authors must have used to measure the spectra (assuming an identical cell was used). The extinction spectra are shown in Figure S.14. Therefore, using Beer-Lambert's Law we can calculate the concentration of photohydrate detected in a film of water with path length of 120 μm as:

$$c = \frac{(2.4 \times 10^{-4})}{(4975 \text{ cm}^{-1}\text{M}^{-1})(0.012 \text{ cm})} = 4.8 \times 10^{-6}\text{M}$$

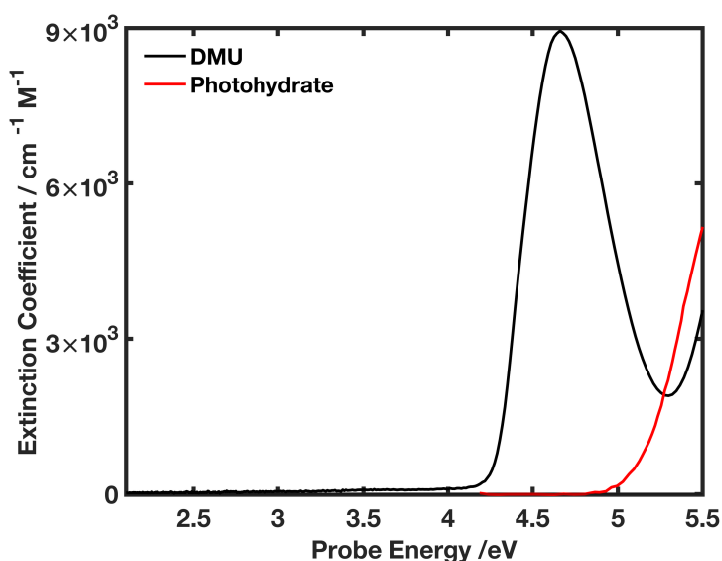


Figure S.14 Extinction spectra of DMU and photohydrate in aqueous solution.²¹

The initial excited state concentration was calculated using the procedure described by van Wilderen *et al.*²² In this calculation, we calculate number of 267 nm photons incident at the sample

$$N_i = \frac{550 \times 10^{-6}\text{J}}{7.44 \times 10^{-19}\text{J}} = 7.39 \times 10^{11} \text{ photons}$$

The peak photon fluence (F) of this pulse with a 170 μm FWHM Gaussian profile is (note s is the standard deviation of the Gaussian-shaped beam)

$$F = \frac{\ln(2) N_i}{2\pi s^2} = 1.56 \times 10^{15} \text{ photons cm}^{-2}$$

With a solution of 10 mM of DMU the absorbance through the film is 1.1, the depth averaged peak fluence is $1.0 \times 10^{15} \text{ photons cm}^{-2}$; yielding an excited state fraction of 0.034 or an excited state concentration of 0.34 mM. Finally, the quantum yield for photohydration is

$$\Phi = \frac{4.0 \times 10^{-6} \text{M}}{3.4 \times 10^{-4} \text{M}} = 1.2 \times 10^{-2}$$

This quantum yield is in excellent agreement with that reported in literature of 1.3×10^{-2} .²³

Alternatively, we can also calculate the quantum yield estimating the initial excited state concentration from the maximum amplitude of the ground state bleach observed at earliest times (we use 0.5 ps, when the contribution of the hot ground state absorption and excited state absorption is minimized). From this approach, we get an absorption of 9.1×10^{-3} at 4.64 eV. Using Beer-Lambert's law we calculated the excited state concentration

$$c = \frac{(9.1 \times 10^{-3})}{(8900 \text{ cm}^{-1} \text{M}^{-1})(0.012 \text{ cm})} = 8.5 \times 10^{-5} \text{M}$$

Therefore, the calculated quantum yield based on this calculated excited concentration and the photohydrate concentration is

$$\Phi = \frac{3.8 \times 10^{-6} \text{M}}{8.5 \times 10^{-5} \text{M}} = 4.5 \times 10^{-2}$$

The discrepancy likely originates from the possibility of excited state absorption still contributing at 0.5 ps, leading to an underestimate of the initial excited state concentration. Regardless, both estimations afford a quantum yield that is within the same order of magnitude as provided by previous steady state measurements. The ability of obtaining the quantum yield of a UV absorbing

photoproduct directly from a time resolved measurement highlights the benefits provided by the use of the soliton probe. In addition, we can state that from our experiments the photohydrate has formed with an upper limit of 70-80 ps (see Figure S12 lower panels).

G. Data Processing

Alongside the standard artifacts that must be corrected for in experiments that utilize white light continua generated in bulk media (e.g. chirp correction), the use of hollow core fibers introduces new issues that must be corrected for.

As mentioned in the main text, there is a coherent artefact that arises from interference of the soliton continuum and light generated at the third harmonic of the same soliton. The centrosymmetric nature of the capillary and the optimized dispersion conditions for soliton formation enable third-harmonic generation (THG), which broadens as the pulse propagates through the hollow-core fiber (Fig. S3). The broadened THG spectrally overlaps with the soliton and interferes with it, producing oscillatory features between 3.5 and 4.1 eV.^{24,25} This interference is highly sensitive to the carrier-envelope phase of the pump laser, which varies from shot to shot, leading to apparent spectral instability. This instability is consequently reflected in the transient absorption measurements. On a timescale of seconds (i.e. comparable with recording averaged spectra at different time delays) the phase of the interference is observed to shift. Without treatment, the wavelengths coincident with the spectral region where interference is manifested become corrupted.

For each transient absorption map we record 10 scans over the time delay. At each time delay the transient absorption is averaged over 500 sequential pairs of pump-on and pump-off shots. The ten scans are stored individually. Using a temporary average over the ten scans, a third order

polynomial is fitted to the center of the two-photon absorption streak as a function of wavelength. This is used to correct the chirp of the white light and line up each of the time delay traces for each wavelength in the 2D map. This is applied to each of the scans individually to allow independent processing of the chirp-corrected scans.

To remove the third harmonic interference contribution, each 2D scan is Fourier transformed along the spectral axis and the high frequency components are suppressed by multiplication with a zero centered Tukey window with an α value of 0.9 and a frequency cut off value of 0.05 pixels^{-1} .²⁶ The 2D map is then inverse Fourier transformed back to pixel space and the scans are averaged together. The results of the filtering process are shown in Figure S.15. The results show how the filtering can substantially remove the spectral oscillations at different probed times while preserving the shape of the main spectral contributions. Since the filtering occurs at every time delay point instead along the whole 2D map at once, it does not affect the dynamics.

We expect that future work will concentrate on approaches to explore phase variations in the interference to robustly remove this artifact, or to shift the phase matching condition to a region of the transient spectrum of less interest to the dynamics being probed. More sophisticated spectral filtering methods are also being explored.

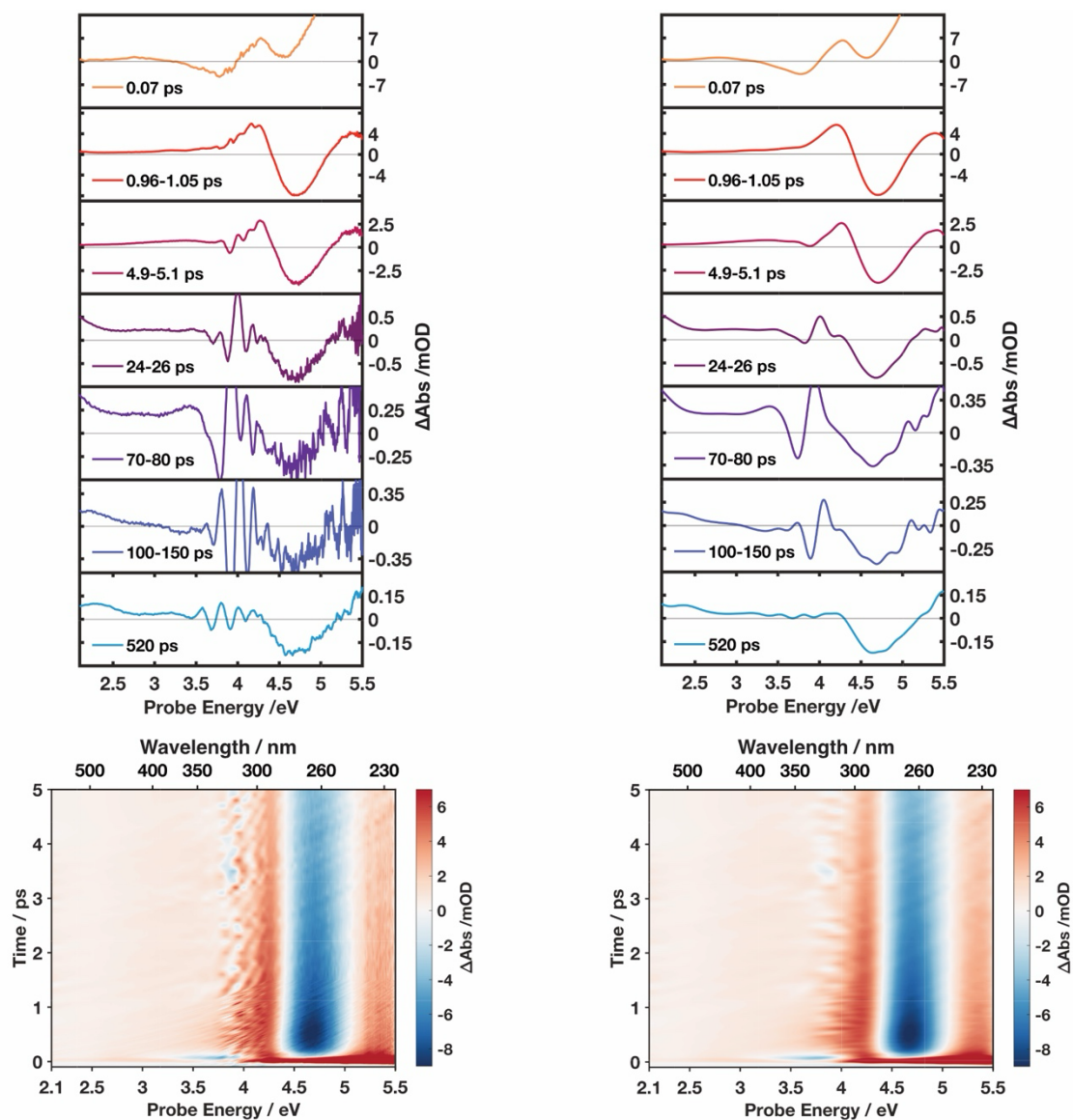


Figure S.15 Spectral slices (top row) and contour plot (bottom row) of the transient absorption dataset before (left) and after (right) filtering.

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